

Cardex Normalization (BCNF)

EXPANSION

Attributes: Set_Code (PK), Set_Abbr (UNIQUE), Set_Name (UNIQUE), Release_Date, Total_Cards

Primary Key: Set_Code

Functional Dependencies:

- Set_Code -> Set_Abbr, Set_Name, Release_Date, Total_Cards
 - Set_Code is primary key which is unique.
- Set_Abbr -> Set_Code, Set_Name, Release_Date, Total_Cards
- Set_Name -> Set_Code, Set_Abbr, Release_Date, Total_Cards
 - Set_Abbr & Set_Name are unique.

Candidate Keys: Set_Code, Set_Abbr, Set_Name

BCNF Verification:

- Set_Code is a superkey and primary key
- Set_Abbr is a superkey
- Set_Name is a superkey

EXPANSION is in BCNF

CARD

Attributes: Card_ID (PK), Set_Code, Card_Name, Card_Number, Category_ID (FK), Rarity_ID (FK)

Primary Key: Card_ID

Functional Dependencies:

- Card_ID -> Set_Code, Card_Name, Card_Number, Category_ID, Rarity_ID
- {Set_Code, Card_Number} -> Card_ID, Card_Name, Category_ID, Rarity_ID

Candidate Keys: Card_ID, {Set_Code, Card_Number}

BCNF Verification:

- Card_ID is a superkey and primary key
- {Set_Code, Card_Number} is a superkey
 - Card numbers are unique within a set

CARD is in BCNF

POKEMON

Attributes: Pokemon_ID (PK), Card_ID (FK), Primary_Type_ID (FK)

Primary Key: Pokemon_ID

Functional Dependencies:

- Pokemon_ID -> Card_ID, Primary_Type_ID
- Card_ID -> Pokemon_ID, Primary_Type_ID

Candidate Keys: Pokemon_ID, Card_ID

BCNF Verification:

- Pokemon_ID is a superkey and primary key
- Card_ID is a superkey

POKEMON is in BCNF

POKEMON_TYPE

Attributes: Type_ID (PK), Type_Name (UNIQUE)

Primary Key: Type_ID

Functional Dependencies:

- Type_ID -> Type_Name
- Type_Name -> Type_ID

Candidate Keys: Type_ID, Type_Name

BCNF Verification:

- Type_ID is superkey and primary key
- Type_Name is a superkey

POKEMON_TYPE is in BCNF

TRAINER

Attributes: Trainer_ID (PK), Card_ID (FK, unique?), Trainer_Subtype_ID (FK)

Primary Key: Trainer_ID

Functional Dependencies:

- Trainer_ID -> Card_ID, Trainer_Subtype_ID
- Card_ID -> Trainer_ID, Trainer_Subtype_ID

Candidate Keys: Trainer_ID, Card_ID

BCNF Verification:

- Trainer_ID is a superkey and primary key
- Card_ID is a superkey

TRAINER is in BCNF

TRAINER_SUBTYPE

Attributes: Trainer_Subtype_ID (PK), Trainer_Subtype_Name (UNIQUE)

Primary Key: Trainer_Subtype_ID

Functional Dependencies:

- Trainer_Subtype_ID -> Trainer_Subtype_Name
- Trainer_Subtype_Name -> Trainer_Subtype_ID

Candidate Keys: Trainer_Subtype_ID, Trainer_Subtype_Name

BCNF Verification:

- Trainer_Subtype_ID is superkey and primary key
- Trainer_Subtype_Name is a superkey

TRAINER_SUBTYPE is in BCNF

CARD_CATEGORY

Attributes: Category_ID (PK), Category_Name (UNIQUE)

Primary Key: Category_ID

Functional Dependencies:

- Category_ID -> Category_Name
- Category_Name -> Category_ID

Candidate Keys: Category_ID, Category_Name

BCNF Verification:

- Category_ID is a superkey and primary key
- Category_Name is a superkey

CARD_CATEGORY is in BCNF

RARITY

Attributes: Rarity_ID (PK), Rarity_Name (UNIQUE)

Primary Key: Rarity_ID

Functional Dependencies:

- Rarity_ID -> Rarity_Name
- Rarity_Name -> Rarity_ID

Candidate Keys: Rarity_ID, Rarity_Name

BCNF Verification:

- Rarity_ID is a superkey and primary key
- Rarity_Name is a superkey

RARITY is in BCNF

VARIANT

Attributes: Variant_ID (PK), Variant_Name (UNIQUE)

Primary Key: Variant_ID

Functional Dependencies:

Variant_ID -> Variant_Name

Variant_Name -> Variant_ID

Candidate Keys: Variant_ID, Variant_Name

BCNF Verification:

- Variant_ID is a superkey and primary key

- Variant_Name is a superkey

VARIANT is in BCNF

COLLECTION

Attributes: Collection_ID (PK), User_ID, Card_ID, Variant_ID, Quantity

Primary Key: Collection_ID

Functional Dependencies:

Collection_ID -> User_ID, Card_ID, Variant_ID, Quantity

{User_ID, Card_ID, Variant_ID} -> Collection_ID, Quantity

Candidate Keys: Collection_ID, {User_ID, Card_ID, Variant_ID}

BCNF Verification:

- Collection_ID is a superkey and primary key

- {User_ID, Card_ID, Variant_ID} is a superkey

COLLECTION is in BCNF

USER

Attributes: User_ID (PK), User_Name (UNIQUE), Password, Email (UNIQUE)

Primary Key: User_ID

Functional Dependencies:

User_ID -> Username, Password, Email

Username -> UserID, Password, Email

Email -> User_ID, Username, Password

Candidate Keys: User_ID, Username, Email

BCNF Verification:

- User_ID is a superkey and primary key

- User_Name is a superkey

- Email is a superkey

USER is in BCNF